

Qatar-1b

R-0631

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230904 · created 2026-07-25T15:52:57+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460191.7157 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.017
Transit depth (Rp/Rs)^2	2.43 +/- 0.52 [%]
Orbital Inclination (inc)	85.27 +/- 1.47
Airmass coefficient 1 (a1)	1.024 +/- 0.014
Airmass coefficient 2 (a2)	-0.02 +/- 0.11
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#3 - [604, 383]
Optimal Aperture	3.71
Optimal Annulus	12.02
Transit Duration (day)	0.0741 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.017 vs 0.1463 (deviation 0.49 σ)
Duration fit vs published	0.0741 d vs 0.0692 d (deviation 0.65 σ)
Mid-time O-C	6.4 min
Depth SNR	7.8
Residual scatter	1.35 %

Light curve

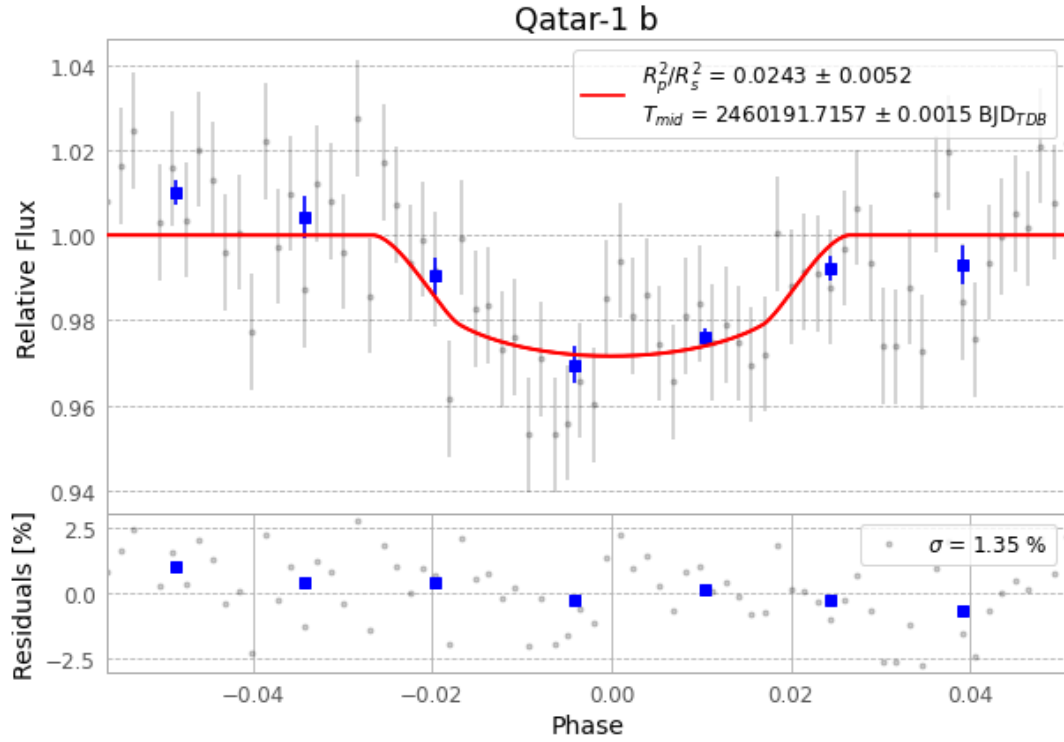


Figure 1 — FinalLightCurve_Qatar-1 b_2023-09-03.png (SHA-256 e39869643f5d6652...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [243, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6982d7335e5c6e4b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1230904031216.FITS → Qatar-1230904065115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230904.log (c481dd61f319...), FinalLightCurve_Qatar-1 b_2023-09-03.png (e39869643f5d...), FinalLightCurve_Qatar-1 b_2023-09-03.csv (cbc0b927fd10...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 15:52Z.